FRAUD DETECTION & RISK ANALYTICS PIPELINE

Presented by Team 3

DOMAIN: BFSI

ENTERPRISE DATA ENGINEERING PIPELINE FOR NOVACRED BANK

Problem Statement

- Processes 2M+ daily transactions (Card, UPI, NEFT, RTGS)
- Enables real-time data processing & fraud detection
- Ensures regulatory compliance (RBI, GDPR)
- Masks sensitive data (PII) before analytics
- Automates SAR reporting & Basel III calculations
- Implements audit, balance & control checks across all layers

Business Problems

01

DELAY

14- hour fraud detection delay

02

Manual SAR Reporting

Manual SAR reporting resulting in (3 days)

03

Excel based calculations

Excel-based capital calculations

04

No unification

Siloed channels → no unified view

05

PII Exposure

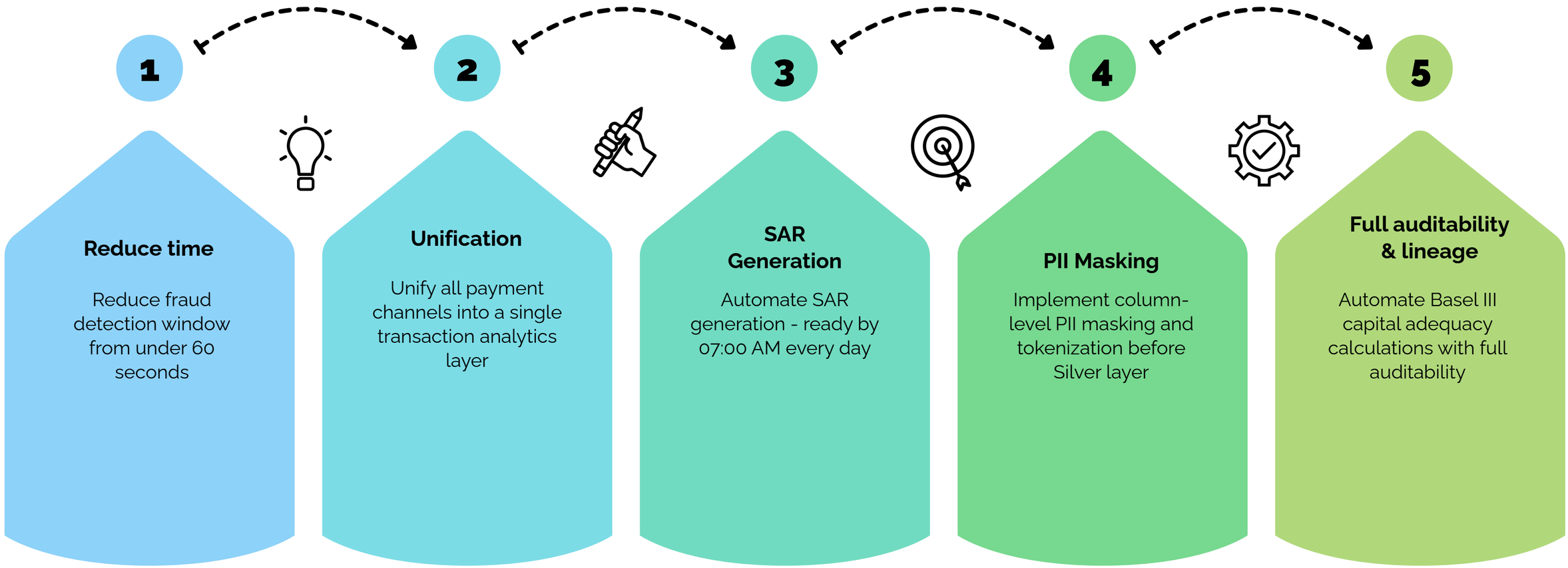
PII Exposure resulting in compliance issues

06

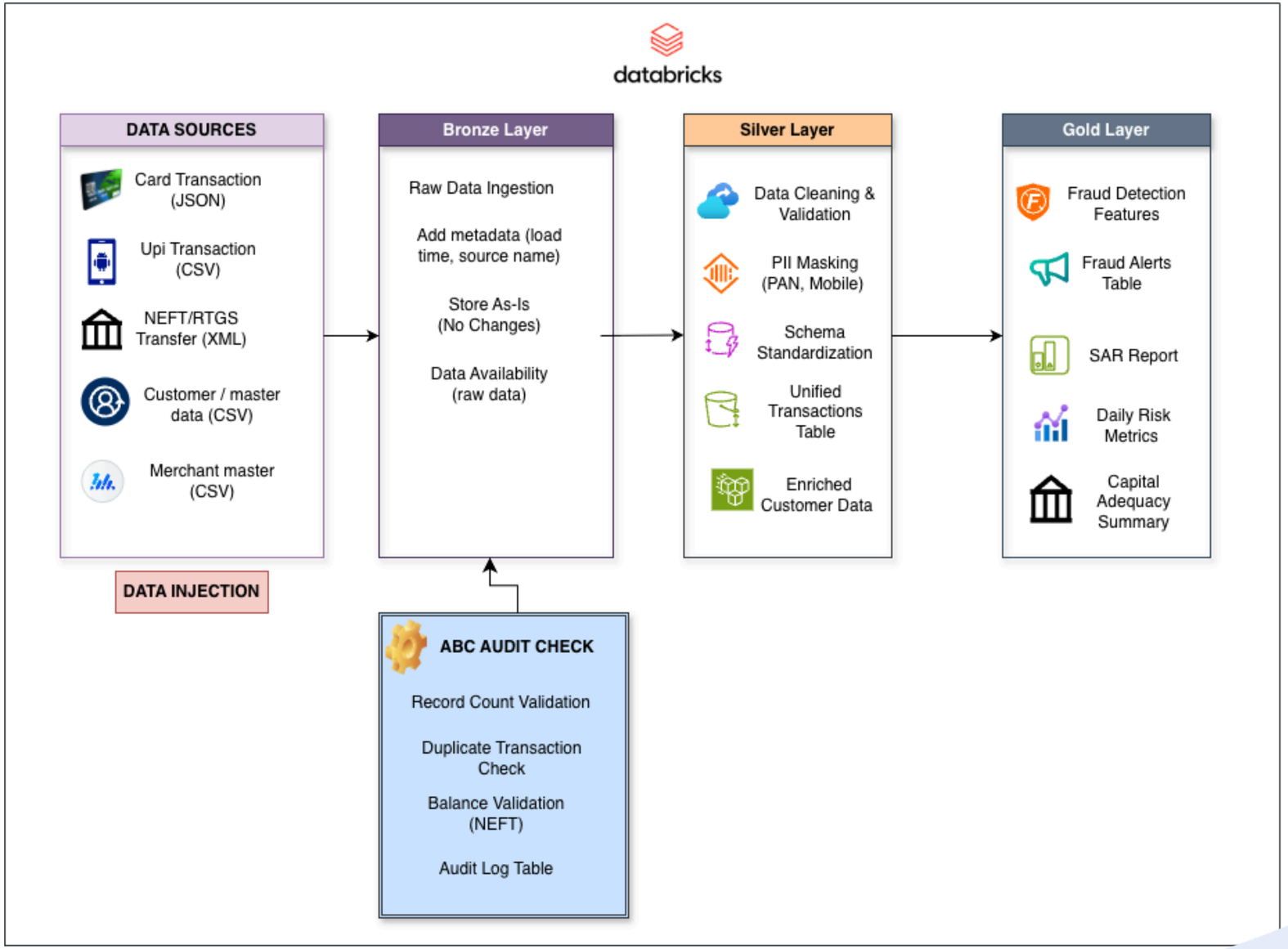
No Audit

No audit traceibility

Target Outcomes



ARCHITECTURE



DATA SOURCE

3.3 NEFT / RTGS Wire Transfers

3.3. neft_rtgs_transactions.xml [XML — ISO 20022 format, SFTP from clearing house]	
Volume	~50K transactions/day
Frequency	Daily SFTP at 23:00 from RBI clearing system
Key Fields	MsgId, CreDtTm, NbOfTxs, TtlIntrBkSttlmAmt, CdtTrfTxInf:[InstrId, Amt, CdtrAcct:IBAN, Dbtr:Nm, Cdtr:Nm, Purp:Cd, RmtInf:Ustrd]
Sample	`<CdtTrfTxInf><InstrId>NEFT001</InstrId><Amt Ccy='INR'>500000.00</Amt><Dbtr><Nm>John Doe</Nm></Dbtr></CdtTrfTxInf>`

DATA SOURCE

3.4 Customer Master Data

3.4. customers_corebanking.csv [CSV — Monthly CDC from core banking]	
Volume	~2M customers
Frequency	Monthly full + daily incremental delta file
Key Fields	customer_id, full_name, pan_number, aadhaar_last4, mobile_number, email, date_of_birth, address, city, state, pincode, account_type, risk_rating (LOW|MEDIUM|HIGH), kyc_status
Sample	`PCI-DSS: pan_number must be tokenized; GDPR: full_name, email, mobile must be masked`

3.5 Merchant Master

3.5. merchant_master.csv [CSV — Reference]	
Volume	~300K merchants
Frequency	Weekly refresh
Key Fields	merchant_id, merchant_name, merchant_category_code, mcc_description, risk_tier (1–5), country, is_blacklisted, registration_date

DATA SOURCE

3.1 Card Transactions

3.1. card_transactions.json [JSON — Real-time Kafka stream + hourly batch]	
Volume	~500K transactions/day
Freque	Real-time Kafka + hourly S3 batch landing
Key Fields	txn_id, card_number (PAN — 16 digits), customer_id, merchant_id, merchant_name, merchant_category_code (MCC), txn_amount, txn_currency, txn_ts, txn_type (PURCHASE|REFUND|WITHDRAWAL), pos_entry_mode, country_code, is_international, auth_code,
Sample	`{"txn_id":"TXN001","card_number":"4111111111111234","customer_id":"CUST001","txn_amount":2500.00,"txn_ts":"2024-01-15T10:30:00","response_code":"00"}`

3.2 UPI Transactions

3.2. upi_transactions.csv [CSV — Pipe-separated, daily batch from NPCI]	
Volume	~1.2M transactions/day
Frequency	Daily batch at 02:00 from NPCI clearing house
Key Fields	upi_txn_id, payer_vpa, payee_vpa, payer_bank, payee_bank, amount, txn_ts, status (SUCCESS|FAILED|PENDING|REVERSED), remarks, device_fingerprint, ip_address, latitude, longitude
Sample	`2024-01-15T10:30:00|PAY001|9876543210@upi|merchant@upi|HDFC|ICICI|5000.00|SUCCESS`

BRONZE LAYER

1

Card Data

Stream data using Auto Loader (STRUCTURED STREAMING)



2

UPI Data

seperated by “|”



5

Merchant Master

CSV





NEFT (National Electronic Funds Transfer) /RTGS(Real-Time Gross Settlement)

3



Customer Master Data

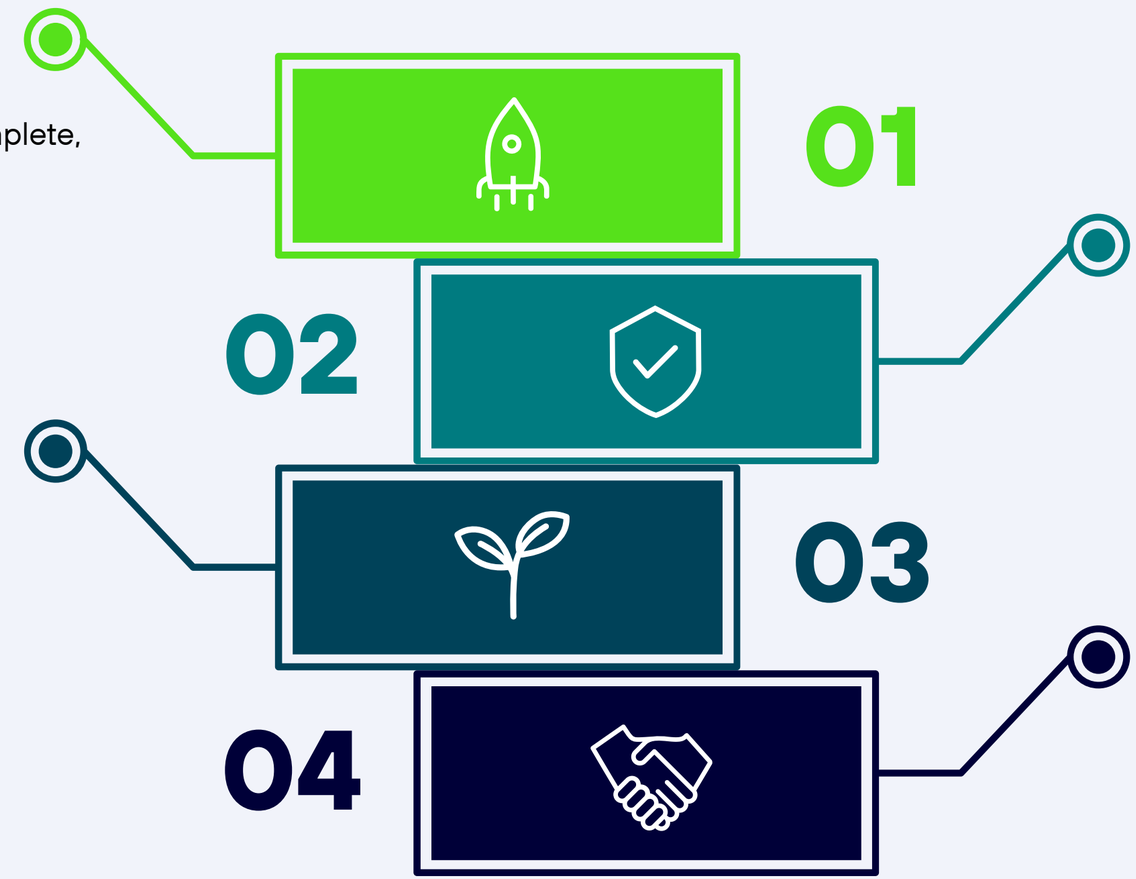
CSV

4

ABC CHECK (Audit – Balance – Control)

What is ABC Check

It is a data quality check
Ensures data is correct, complete, and reliable



A – Audit Check

- Check total number of records
- Make sure no data is missing
- Example:
- Card transactions should be within expected range

B – Balance Check

- Check if total amounts are correct
- Ensure no mismatch in money values

C – Control Check

- Check for duplicate transactions
- Ensure no repeated or wrong entries

SILVER LAYER

What is Personally Identifiable Information (PII)?

Personally Identifiable Information (PII) refers to sensitive user data that can be used to identify an individual. This includes information such as:

- Permanent Account Number (PAN)
- Mobile Number
- Email Address



Problem

- The raw data includes sensitive customer information.
- Exposure to this data can result in:
  - Data breaches
  - Identity theft

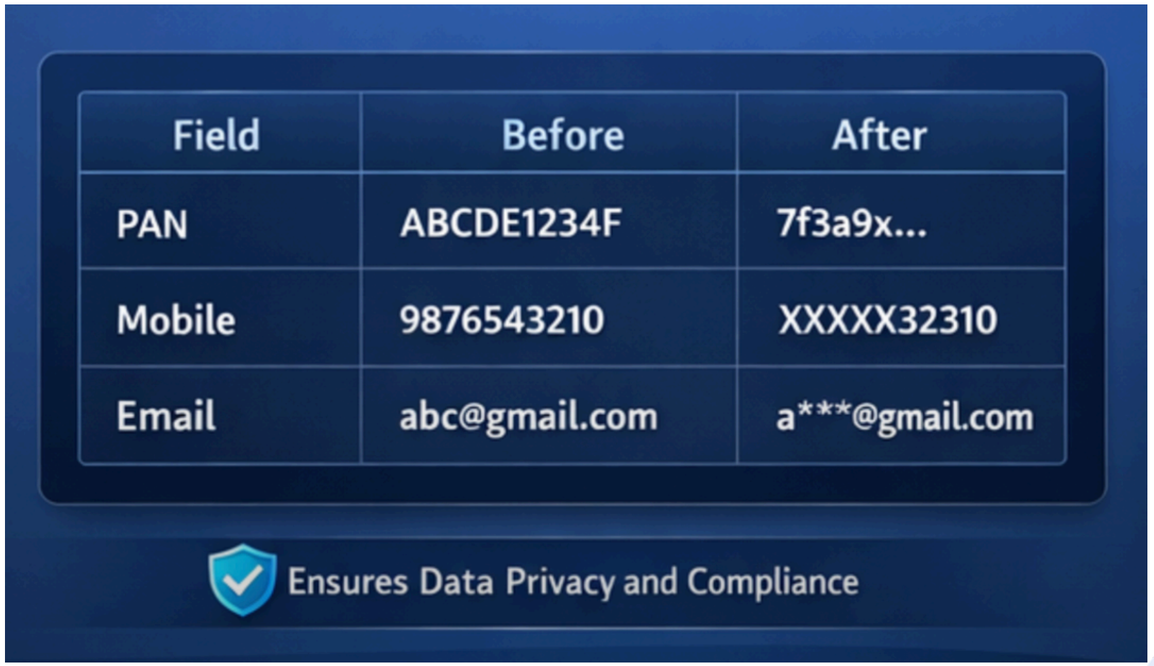
PII MASKING IMPLEMENTATION

Techniques Used

- PAN → Hashed using SHA-256
- Mobile → Partial masking (XXXXXX1234)
- Email → Masked (a***@gmail.com)

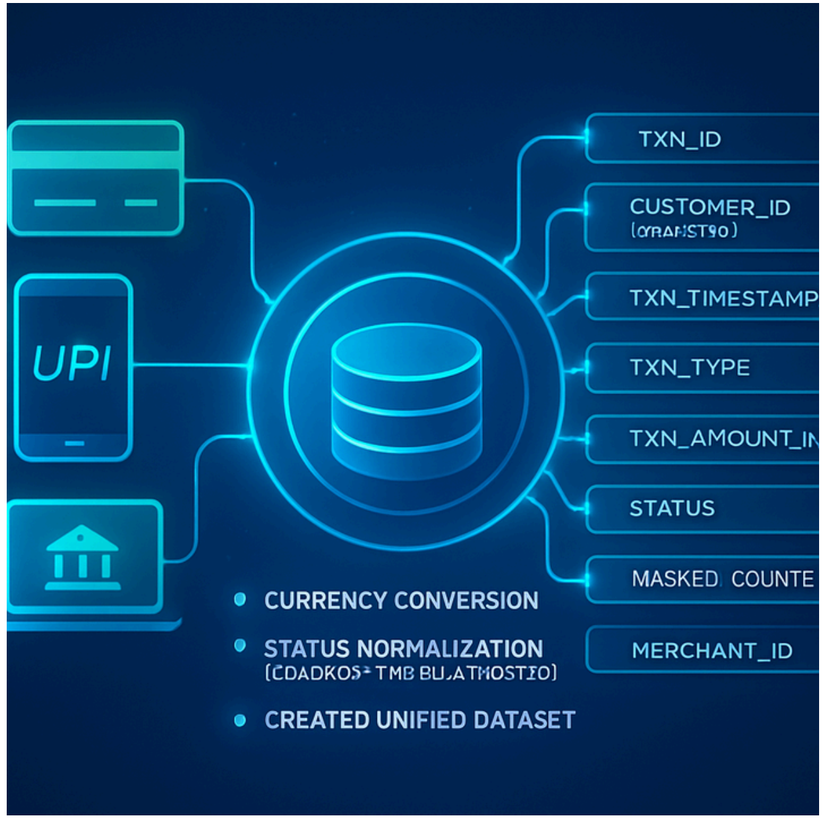
Pipeline Integration

- Applied in Silver Layer
- Ensures:
  - No raw PII in Gold layer
  - Secure downstream analytics



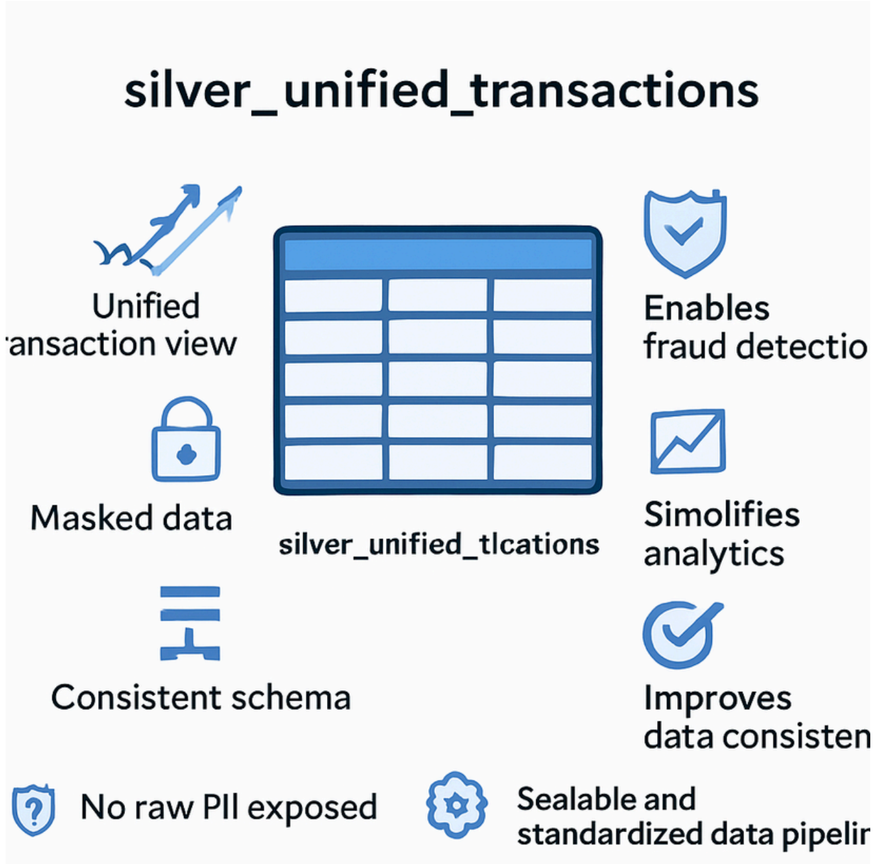
Unified Transaction Layer

- **INTEGRATED MULTIPLE TRANSACTION SOURCES:**
  CARD, UPI, NEFT/RTGS.
- **STANDARDIZED SCHEMA ACROSS ALL SYSTEMS:**
  TXN_ID, CUSTOMER_ID (MASKED), TXN_TIMESTAMP
  TXN_TYPE, TXN_AMOUNT_INR, STATUS
  MASKED_COUNTERPARTY, MERCHANT_ID,
  IS_INTERNATIONAL
- **PERFORMED KEY TRANSFORMATIONS:**
  CURRENCY CONVERSION to INR
  STATUS NORMALIZATION (SUCCESS, FAILED,PENDING)
  ADDED TXN_TYPE, CHANNEL, PAYMENT_MODE
- **CREATED UNIFIED DATASET USING UNIONBYNAME()**



Final Output & Impact

- **Generated final table:**
  silver_unified_transactions
- **Key features:**
  Unified transaction view across all channels
  Masked customer and counterparty data
  Consistent schema for downstream use
- **Benefits:**
  Enables fraud detection
  Simplifies analytics across systems
  Improves data consistency and usability
- **Ensured:**
  No raw PII exposed
  Scalable and standardized data pipeline



Fraud Feature Engineering

Turning raw transactions into meaningful patterns for fraud detection

Transaction Count (1hr)

- Number of transactions in last hour
- Detects high activity

Transaction Amount (1hr)

- Total spend in last one hour
- Detects sudden spending spikes

Distinct Merchants (1hr)

- Unique merchants in one hour
- Shows unusual merchants behaviour

Distinct Countries (24h)

- Countries used in last 24 hours
- Helps detect impossible travel

Time Since Last Txn

- Gap between transactions
- Detects rapid activity

Amount Z-Score

- Deviation from normal spending
- Detects unusual transaction

RULES-BASED FRAUD DETECTION

Detecting suspicious transactions using simple rules

Velocity Breach

Too many transactions (>10 in 1 hour)

Behavior change

Amount > 10× customer average

Multi Rule Trigger

One transaction can trigger multiple rules



High Spending

Amount exceeds ₹500K in 1 hour

Geo Anomaly

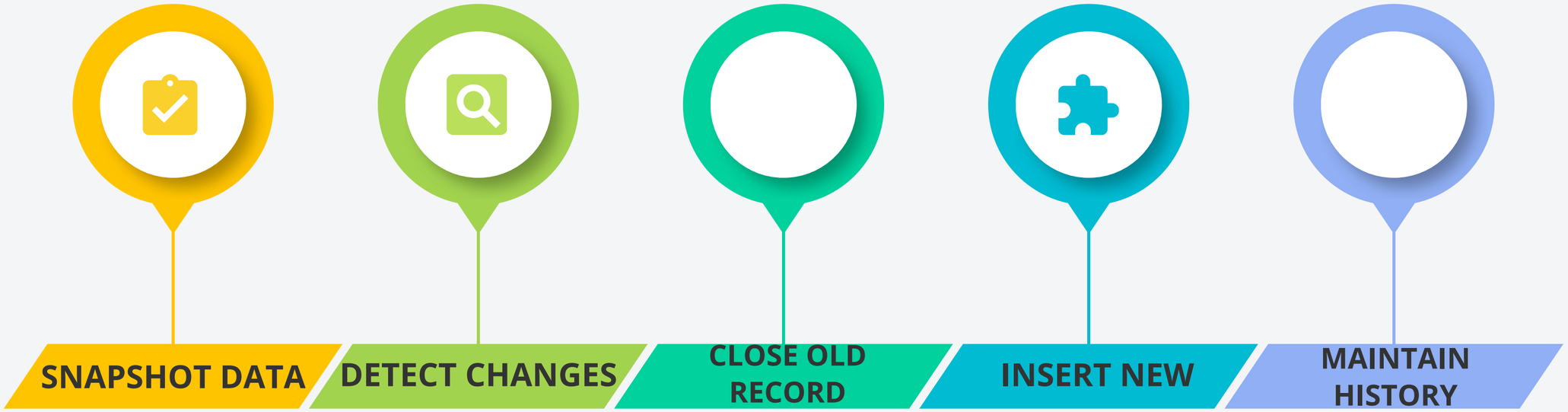
Multiple countries within 24 hours (>3)

International Risk

Foreign transaction with high amount (>₹100K)

SCD_2 – Customer Risk Tracking

Maintaining historical changes in customer risk for audit and compliance



Retrieve old and new customer data snapshots

Identify changes in risk, KYC, or account type

Set end_date and mark is_current = false

Add new values and mark is_current = true

Track using start_date and end_date

```
display(df_snapshot1)
```

See performance (1)

Table

	customer_id	risk_rating	kyc_status	account_type
1	CUST0001858	LOW	VERIFIED	SAVINGS
2	CUST0000703	LOW	VERIFIED	SAVINGS
3	CUST0000169	LOW	VERIFIED	SAVINGS
4	CUST0001940	LOW	VERIFIED	SAVINGS
5	CUST0000680	LOW	VERIFIED	SAVINGS

Snapshot 1

- old data

```
display(df_snapshot2)
```

See performance (1)

Table

	customer_id	risk_rating	kyc_status	account_type
1	CUST0001858	HIGH	VERIFIED	SAVINGS
2	CUST0000703	LOW	PENDING	SAVINGS
3	CUST0000169	LOW	VERIFIED	SAVINGS
4	CUST0001940	LOW	VERIFIED	SAVINGS
5	CUST0000680	LOW	VERIFIED	SAVINGS

Snapshot 2

- new data

```
df = spark.table("`bfsi-project-team-3`.`silver-tables`.dim_customer_risk")

display(df.orderBy("customer_id", "effective_start_date"))
```

Table

Terminal (+2)

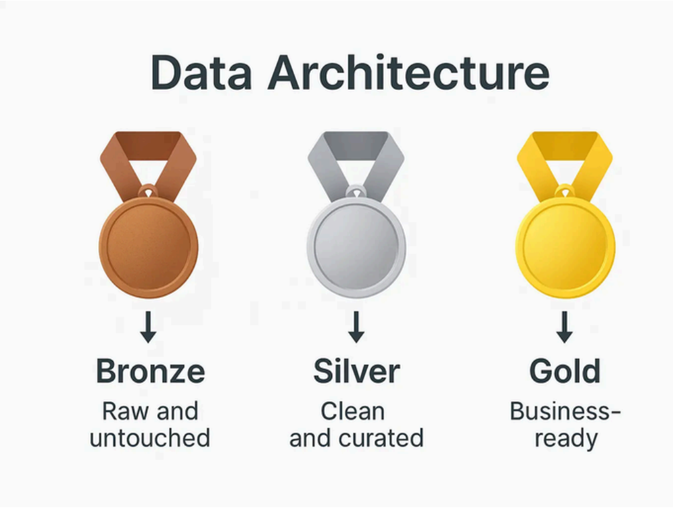
	customer_id	risk_rating	kyc_status	account_type	effective_start_date	effective_end_date	is_current
1	CUST0000169	LOW	VERIFIED	SAVINGS	2026-03-26T18:52:47.765+00:00	null	true
2	CUST0000680	LOW	VERIFIED	SAVINGS	2026-03-26T18:52:47.765+00:00	null	true
3	CUST0000703	LOW	VERIFIED	SAVINGS	2026-03-26T18:52:47.765+00:00	2026-03-26T18:54:06.860+00:00	false
4	CUST0000703	LOW	PENDING	SAVINGS	2026-03-26T18:59:29.100+00:00	null	true
5	CUST0001858	LOW	VERIFIED	SAVINGS	2026-03-26T18:52:47.765+00:00	2026-03-26T18:54:06.860+00:00	false
6	CUST0001858	HIGH	VERIFIED	SAVINGS	2026-03-26T18:59:29.100+00:00	null	true
7	CUST0001940	LOW	VERIFIED	SAVINGS	2026-03-26T18:52:47.765+00:00	null	true

Final SCD-2 Table

- merged both table
- history maintained

Gold Layer

The Gold Layer represents the final stage of the data pipeline where processed data is transformed into business-ready insights.



In this project, the Gold Layer produces three key outputs:

- Suspicious Activity Reports (SAR) for regulatory compliance
- Daily Risk Metrics for monitoring fraud trends
- Capital Adequacy calculations based on Basel III norms

Suspicious Activity Report (SAR)

The SAR report identifies transactions that may indicate fraudulent or suspicious behavior and prepares them for regulatory reporting (RBI compliance).

Transactions are flagged based on:

- High-value transactions exceeding ₹10,00,000 in a single day per customer
- Transactions already marked as fraudulent using rule-based detection



Daily Risk Metrics

This table provides a comprehensive overview of daily transaction activity and fraud trends, enabling risk teams to monitor system health and detect anomalies.

Key metrics include:

- Total number of transactions and total transaction value
- Number and value of fraud-flagged transactions
- Fraud rate percentage calculated as : Fraud Rate = Fraud Transactions / Total Transactions

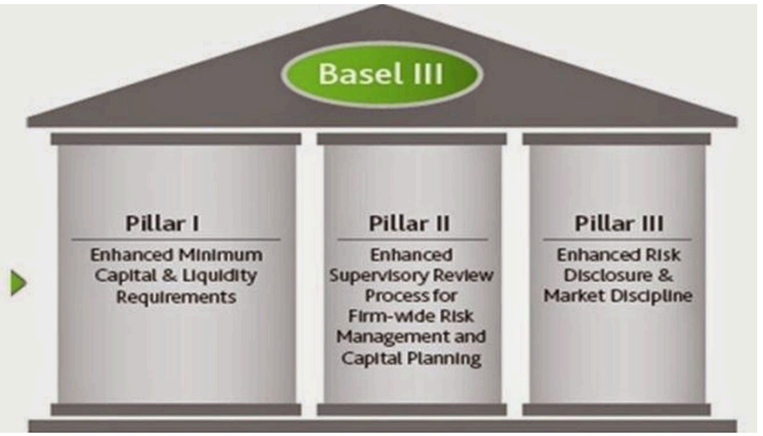
Capital Adequacy Calculation (Basel III)

This module evaluates the financial stability of the bank using Basel III guidelines by calculating Risk-Weighted Assets (RWA) and the Capital Adequacy Ratio (CAR).

The calculation follows:

RWA = EAD × LGD × PD × 12.5

CAR = (Tier 1 Capital + Tier 2 Capital) / Total RWA × 100



- EAD (Exposure at Default) is approximated using transaction amounts
- LGD (Loss Given Default) is assigned based on transaction risk (international vs domestic)
- PD (Probability of Default) is derived from fraud indicators

Airflow:

bfsi_daily_risk_pipeline with SLA Enforcement

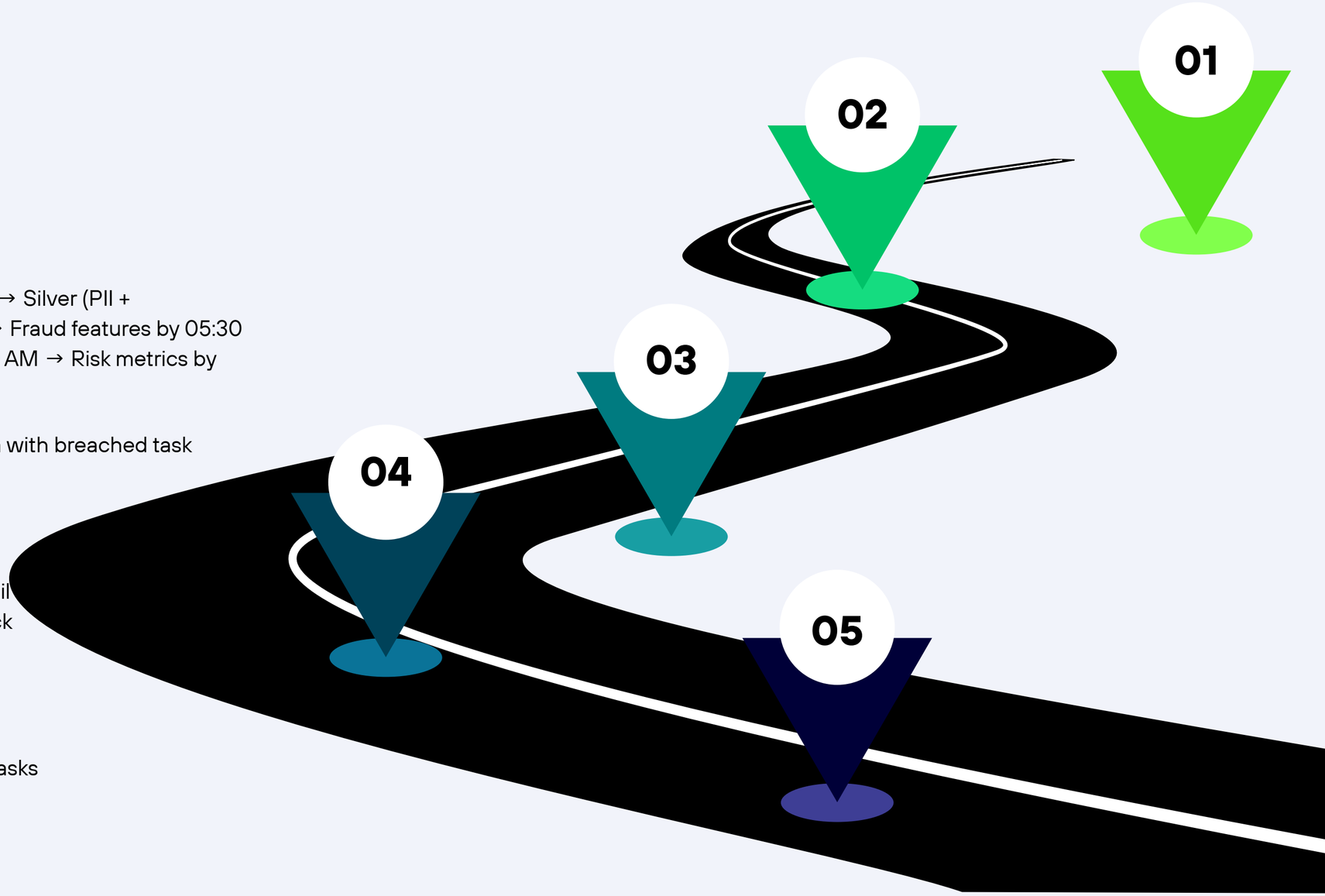
Schedule — 0 1 * * * (1:00 AM daily)

SLA Chain — Bronze done by 04:00 AM → Silver (PII + unification) by 05:00 AM → Fraud features by 05:30 AM → SAR report by 06:50 AM → Risk metrics by 07:00 AM

SLA Breach — send email to risk-ops team with breached task name and delay duration

ABC Check Task — BranchPythonOperator - fail entire DAG if any ABC check fails

Retry Policy — Retries with 5-minute exponential backoff for all tasks



Conclusion

End-to-end pipeline

Successfully built an end-to-end data pipeline from ingestion to final reporting

01

Multiple data formats processed

Processed multiple data formats including JSON, CSV, and XML

02

Data quality checks

Implemented strong data quality checks using Audit, Balance, and Control framework

03

Data compliance, audits

Ensured data compliance, traceability, and audit logging across all layers

04

SAR Reports and risk metrics

Delivered meaningful outputs like fraud alerts, SAR reports, and risk metrics

05

THANK YOU